

DR ESLAM AHMED
eliteigcse.com

Transformations of Functions

Inside the bracket moves x — the opposite way.
Outside moves y — the way you see. Move the key points, then draw.

THE TOPIC ON ONE MAP

Transformations of Functions

01 — Map a point

$f(x + 3)$ $f(3x)$ find a

02 — Sketch the transformed curve

asymptote touches the axis y -intercept

03 — Transformed curves meeting

$f(2x) = f(x) + 3$ $f(x) < 6$

$f(x) + a$	translate $\begin{pmatrix} 0 \\ a \end{pmatrix}$	$y + a$
$f(x + a)$	translate $\begin{pmatrix} -a \\ 0 \end{pmatrix}$	$x - a$ (opposite)
$af(x)$	stretch y -axis, s.f. a	$y \times a$
$f(ax)$	stretch x -axis, s.f. $\frac{1}{a}$	$x \div a$ (opposite)
$-f(x)$	reflect in the x -axis	$y \rightarrow -y$
$f(-x)$	reflect in the y -axis	$x \rightarrow -x$

01 Map a Point

Inside changes x the opposite way. Outside changes y the same way. Never both.

WHEN YOU SEE → YOU DO

“state the coordinates of the image of P ”

→ inside → x , outside → y

“ $f(x + 3)$ ”

→ $x - 3$ (opposite)

“ $f(3x)$ ”

→ $x \div 3$

“ P is mapped to P' , find a ”

→ compare the coordinates, solve

“describe fully the transformation”

→ name + vector / scale factor + direction

WORKED MODEL • OCTOBER 2025

$P\left(\frac{3\pi}{2}, 4\right)$ on $y = f(x)$.

1 $f\left(x - \frac{\pi}{6}\right): x + \frac{\pi}{6}$

inside, opposite

2 $-\frac{1}{2}f(x): y \times \left(-\frac{1}{2}\right)$

outside, same

ANSWER $\left(\frac{5\pi}{3}, 4\right), \left(\frac{3\pi}{2}, -2\right)$

IDEA 01

Practise: Map a Point

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Inside or outside?

P1 · TOPIC 7 4 marks

$P(4, -2)$ lies on $y = f(x)$. Find the image of P on (a) $y = f(x) + 5$ (b) $y = f(x - 3)$
(c) $y = 2f(x)$ (d) $y = f(2x)$

Q2 DR ESLAM Describe fully

P1 · TOPIC 7 2 marks

Describe the transformation that maps $y = x^2$ onto $y = (x + 5)^2$.

SELECTED PRACTICE

Q3**PRACTICE**

Practice · Easy

P1 · TOPIC 7 3 marks

3 The point $P(2,6)$ lies on the curve with equation $y = f(x)$.

State the coordinates of the image of point P on the curves with the following equations:

(i) $y = f(x) + 1$

(ii) $y = -f(x)$

(iii) $y = f\left(\frac{1}{4}x\right)$.

(3 marks)

Q4**PRACTICE**

Practice · Medium

P1 · TOPIC 7**4 marks**

- 1 The point $P(-1, 4)$ lies on the curve with equation $y = f(x)$.

State the coordinates of the image of point P on the curves with the following equations:

(i) $y = f(x) + 3$

(ii) $y = f(x + 3)$

(iii) $y = 3f(x)$

(iv) $y = f(3x)$

(4 marks)

Q5**PRACTICE** Practice · Medium**P1 · TOPIC 7 4 marks**

3 The point $P(3,2)$ lies on the curve with equation $y = f(x)$.

(i) On the graph of $y = f(x) + a$, where a is a constant, the point P is mapped to the point $(3, -5)$. Determine the value of a .

(ii) On the graph of $y = f(x + b)$, where b is a constant, the point P is mapped to the point $(-1, 2)$. Determine the value of b .

(iii) On the graph of $y = cf(x)$, where c is a constant, the point P is mapped to the point $(3, 1)$. Determine the value of c .

(iv) On the graph of $y = f(dx)$, where d is a constant, the point P is mapped to the point $(1, 2)$. Determine the value of d .

(4 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA11 · January 2023 · Q7

P1 · TOPIC 7 10 marks

7. (a) On Diagram 1, sketch a graph of the curve
- C
- with equation

$$y = \frac{6}{x} \quad x \neq 0 \quad (2)$$

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \quad x \neq 2$

- (b) Fully describe this transformation.
- (2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .Given that the x coordinate of point P is -4

- (c) find the value of
- k
- ,
- (2)

- (d) find, using algebra, the coordinates of point
- Q
- .

(Solutions relying entirely on calculator technology are not acceptable.)(4)

Question 7 continued

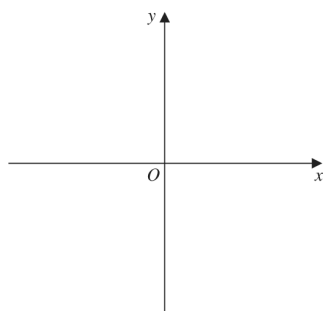
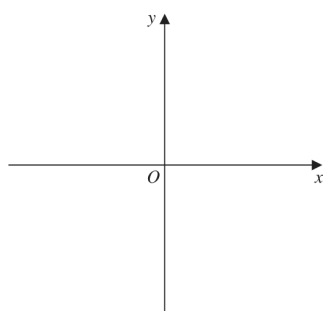


Diagram 1

Only use this copy of Diagram 1 if you need to redraw your graph.



Copy of Diagram 1

(Total for Question 7 is 10 marks)

Q7**PAST PAPER**

Pearson IAL WMA11 · October 2025 · Q9

P1 · TOPIC 7 6 marks

9.

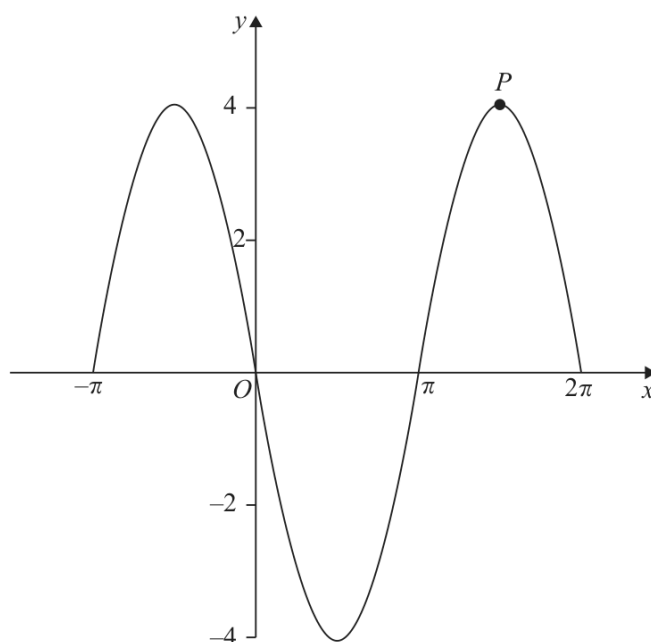
**Figure 3**

Figure 3 shows a sketch of part of the graph of the trigonometric function with equation $y = f(x)$

- (a) Write down an expression for $f(x)$

(2)

The point P lies on $y = f(x)$ and is shown in Figure 3.

- (b) State the coordinates of the point to which P is transformed when the graph of $y = f(x)$ is transformed to the graph with equation

(i) $y = f\left(x - \frac{\pi}{6}\right)$

(2)

(ii) $y = -\frac{1}{2}f(x)$

(2)**(Total for Question 9 is 6 marks)**

02 Sketch the Transformed Curve

List the key points — roots, y-intercept, turning points, asymptotes — move each one, then draw the same shape through them.

WHEN YOU SEE → YOU DO

**“on separate diagrams sketch
 $y = f(x + 3)$ and $y = f(-3x)$ ”**

→ move every key point

“state the equation of the asymptote”

→ move the asymptote too

“touches the x-axis”

→ a touching point stays a touching point

y-intercept after $f(x + a)$

→ it is a new point: $f(a)$

WORKED MODEL · JUNE 2022

Min $P(-1, 0)$, max $Q\left(\frac{3}{2}, 2\right)$, asymptote $y = 1$.

1 $f(x) - 2$: $(-1, -2)$, $\left(\frac{3}{2}, 0\right)$, $y = -1$

down 2

2 $f(-x)$: $(1, 0)$, $\left(-\frac{3}{2}, 2\right)$, $y = 1$

reflect in y-axis

ANSWER same shape, points moved

IDEA 02

Practise: Sketch the Transformed Curve

My questions first, then selected practice, then real past papers.

Q8

DR ESLAM

Move the key points

P1 · TOPIC 7

3 marks

$y = f(x)$ has a maximum at $(1, 4)$ and an asymptote $y = 2$. State the maximum and the asymptote of $y = f(x + 2) - 1$.

SELECTED PRACTICE

Q9

PRACTICE Practice · Easy

P1 · TOPIC 7 7 marks

6 (a) The function $f(x)$ is defined as $f(x) = (x - 2)(x - 6)$

Sketch the graph of $y = f(x)$, , showing clearly the coordinates of the points where the graph intersects the coordinate axes and the coordinates of the turning point.

(3 marks)

(b) On separate diagrams sketch the graphs of:

(i) $y = f(x - 4)$,

(ii) $y = f(-x)$.

In each case clearly show the coordinates of the points where the graph intersects the coordinate axes and the coordinates of the turning point.

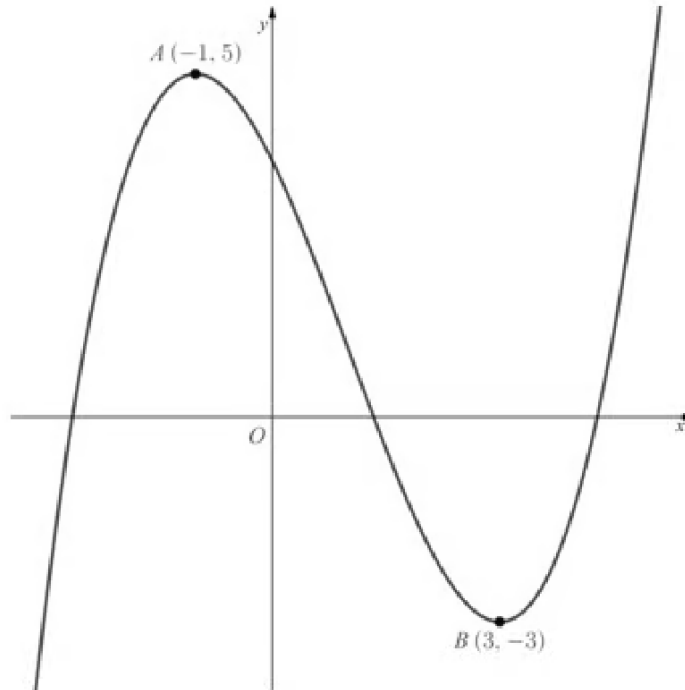
(4 marks)

Q10**PRACTICE**

Practice · Medium

P1 · TOPIC 7 6 marks

- 4 (a) The diagram below shows the graph of $y = f(x)$. The two marked points $A(-1, 5)$ and $B(3, -3)$ lie on the graph.



In separate diagrams, sketch the curves with equation

(i) $y = f(x - 1)$

(ii) $y = f(x) + 3$

On each diagram, give the coordinates of the images of points A and B under the given transformation.

(4 marks)

- (b) On the graph of $y = f(x + a)$ the image of one of the two marked points has an x coordinate of 2. Find the two possible values of a .

(2 marks)

PAST PAPERS

Q11 PAST PAPER Pearson IAL WMA11 · June 2022 · Q4

P1 · TOPIC 7 6 marks

4.

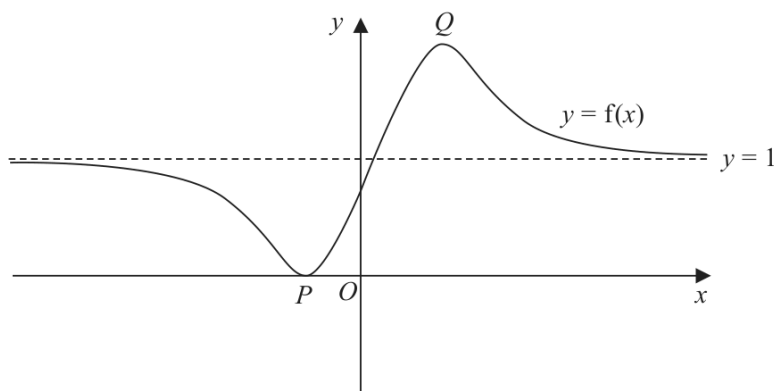


Figure 1

Figure 1 shows a sketch of a curve with equation $y = f(x)$

The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

The line with equation $y = 1$ is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i) $y = f(x) - 2$ (3)

(ii) $y = f(-x)$ (3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

Q12**PAST PAPER**

Pearson IAL WMA11 · June 2024 · Q3

P1 · TOPIC 7 6 marks

3.

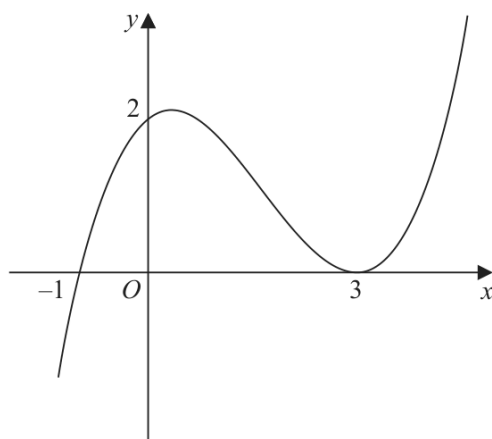
**Figure 1**

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

The curve passes through the points $(-1, 0)$ and $(0, 2)$ and touches the x -axis at the point $(3, 0)$.

On separate diagrams, sketch the curve with equation

(a) $y = f(x + 3)$

(3)

(b) $y = f(-3x)$

(3)

On each diagram, show clearly the coordinates of all the points where the curve cuts or touches the coordinate axes.

(Total for Question 3 is 6 marks)

03 Transformed Curves Meeting

Two transformed curves meet where their equations are equal. Write each one as an equation in x , then solve.

WHEN YOU SEE → YOU DO

“ $y = f(2x)$ meets $y = f(x) + 3$ ”

$$\rightarrow \sqrt{2x} = \sqrt{x} + 3$$

“show that the x coordinate satisfies $\sqrt{x} = \dots$ ”

→ collect $\sqrt{x}$, factorise

“in simplest form”

→ rationalise, then square

“find the range for which $f(x) < 6$ from the graph”

→ read where the curve is below $y = 6$

WORKED MODEL • OCTOBER 2021

$f(x) = \sqrt{x}$; $y = f(2x)$ meets $y = f(x) + 3$.

$$1 \quad \sqrt{2}\sqrt{x} = \sqrt{x} + 3 \Rightarrow \sqrt{x}(\sqrt{2} - 1) = 3$$

collect, factorise

$$2 \quad \sqrt{x} = \frac{3}{\sqrt{2} - 1} = 3(\sqrt{2} + 1)$$

rationalise

ANSWER $Q(27 + 18\sqrt{2}, 6 + 3\sqrt{2})$

IDEA 03

Practise: Transformed Curves Meeting

My questions first, then real past papers.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WMA11 · October 2021 · Q9

P1 · TOPIC 7 9 marks

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

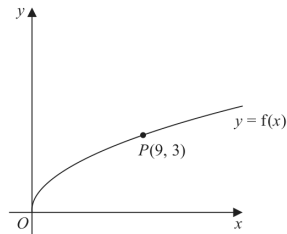


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(3)

(c) Hence find, in simplest form, the coordinates of Q .

(3)

Question 9 continued

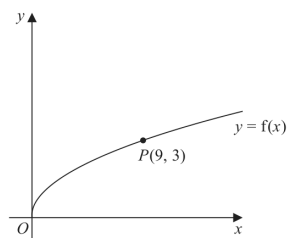
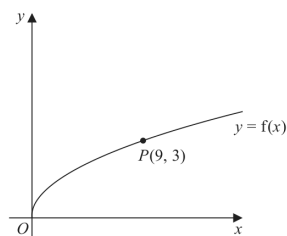


Diagram 1

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

Only use this copy if you need to redraw your graphs.



Copy of Diagram 1

(Total 9 marks)

Q14 PAST PAPER Pearson IAL WMA11 · October 2022 · Q7

P1 · TOPIC 7 8 marks

7.

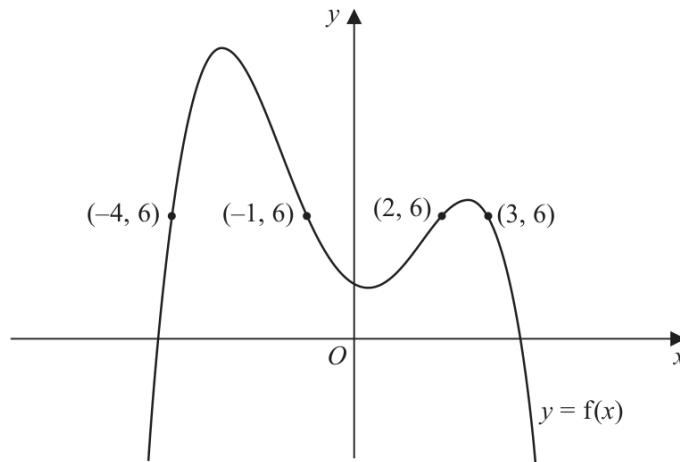
**Figure 1**

Figure 1 shows the curve with equation $y = f(x)$.

The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6 \quad (3)$$

(b) State the largest solution of the equation

$$f(2x) = 6 \quad (1)$$

(c) (i) Sketch the curve with equation $y = f(-x)$.

On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

(ii) Hence find the set of values of x for which

$$f(-x) \geq 6 \text{ and } x < 0 \quad (4)$$

(Total for Question 7 is 8 marks)

9.

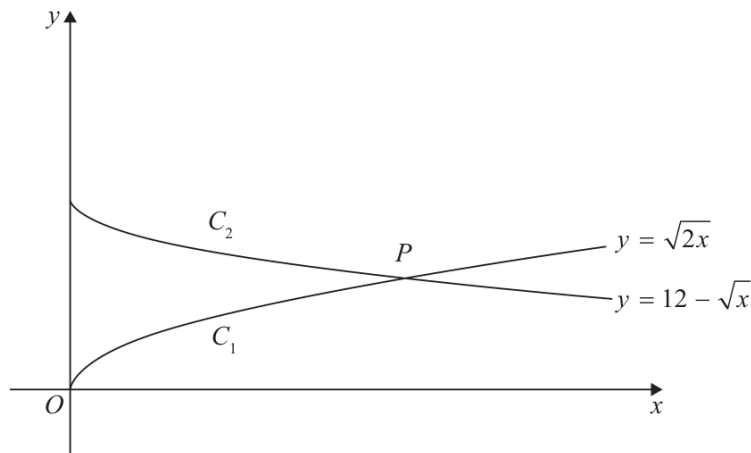


Figure 4

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Figure 4 shows a sketch of

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

(a) Describe fully the single transformation that would transform

- the graph with equation $y = \sqrt{x}$ onto C_1
- the graph with equation $y = -\sqrt{x}$ onto C_2

(4)

The graphs C_1 and C_2 meet at the point P , as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12(\sqrt{2} - 1)$$

(ii) Hence find, in simplest form, the exact coordinates of P .

(6)

(Total for Question 9 is 10 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|--------------------|--|
| 01 | Map a point | Inside $\rightarrow$ x, opposite. Outside $\rightarrow$ y, same. |
| 02 | Sketch | Move every key point and asymptote; keep the shape. |
| 03 | Meeting | Write both as equations, set equal, solve exactly. |

BOTTOM LINE Inside the bracket: x, the opposite way · outside: y, the way you see $\rightarrow$ move the key points.

Answers

Transformations of Functions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • MAP A POINT

- Q1** Dr Eslam • Inside or outside? (a) (4, 3) (b) (7, -2) (c) (4, -4) (d) (2, -2)
- Q2** Dr Eslam • Describe fully translation by $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$
- Q3** Practice • Easy (i) (2, 7) (ii) (2, -6) (iii) (8, 6)
- Q4** Practice • Medium (i) (-1, 7) (ii) (-4, 4) (iii) (-1, 12) (iv) $\left(-\frac{1}{3}, 4\right)$
- Q5** Practice • Medium $a = -7, b = -4, c = \frac{1}{2}, d = 3$
- Q6** Pearson IAL WMA11 • January 2023 • Q7 (b) translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (c) $k = 2$ (d) $Q\left(\frac{5}{2}, 12\right)$
- Q7** Pearson IAL WMA11 • October 2025 • Q9 (a) $f(x) = -4 \sin x$ (b)(i) $\left(\frac{5\pi}{3}, 4\right)$ (ii) $\left(\frac{3\pi}{2}, -2\right)$

IDEA 02 • SKETCH THE TRANSFORMED CURVE

- Q8** Dr Eslam • Move the key points maximum (-1, 3), asymptote $y = 1$
- Q9** Practice • Easy (a) roots 2, 6, y-int 12, min (4, -4) (b)(i) roots 6, 10, y-int 60, min (8, -4) (ii) roots -2, -6, y-int 12, min (-4, -4)
- Q10** Practice • Medium (a)(i) A(0, 5), B(4, -3) (ii) A(-1, 8), B(3, 0) (b) $a = -3$ or $a = -1$
- Q11** Pearson IAL WMA11 • June 2022 • Q4 (i) min (-1, -2), max $\left(\frac{3}{2}, 0\right)$, $y = -1$ (ii) min (1, 0), max $\left(-\frac{3}{2}, 2\right)$, $y = 1$
- Q12** Pearson IAL WMA11 • June 2024 • Q3 (a) cuts (-4, 0), passes (-3, 2), touches (0, 0) (b) cuts $\left(\frac{1}{3}, 0\right)$, touches (-1, 0), y-int (0, 2)

IDEA 03 · TRANSFORMED CURVES MEETING

- Q13** Pearson IAL WMA11 · October 2021 · Q9 (a) $(4.5, 3)$ and $(9, 6)$ (c) $Q(27 + 18\sqrt{2}, 6 + 3\sqrt{2})$
- Q14** Pearson IAL WMA11 · October 2022 · Q7 (a) $x < -4, -1 < x < 2, x > 3$ (b) $x = \frac{3}{2}$ (c)(i) $(4, 6), (1, 6), (-2, 6), (-3, 6)$ (ii) $-3 \leq x \leq -2$
- Q15** Pearson IAL WMA11 · June 2025 · Q9 (a)(i) stretch $\parallel$ x-axis s.f. $\frac{1}{2}$ (ii) translation $\begin{pmatrix} 0 \\ 12 \end{pmatrix}$ (b)(ii) $P(432 - 288\sqrt{2}, 24 - 12\sqrt{2})$